

Welcome to Code Crunch!

C[6] | [CYBER CRUNCH: Cryptography Basics]

[Thu 1-2PM] Weekly | Spring 2025

Attendance

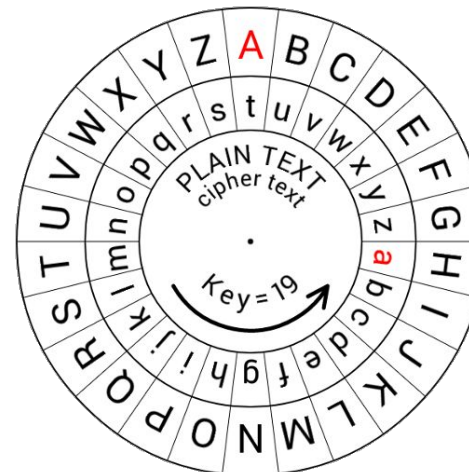


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What is Cryptography?

- **Definition:** The science of encoding and decoding messages to protect data.
- **Key Objectives:**
 - **Confidentiality:** Ensuring data is private.
 - **Integrity:** Ensuring data hasn't been altered.
 - **Authentication:** Verifying identities.
- **Historical Context:** From Caesar ciphers to modern encryption algorithms.





Types of Cryptography

- **Symmetric Key Cryptography:**
 - Uses the same key for encryption and decryption.
 - Example: AES (Advanced Encryption Standard).
- **Asymmetric Key Cryptography:**
 - Uses a pair of keys (public and private).
 - Example: RSA (Rivest–Shamir–Adleman).
- **Hashing:**
 - Converts data into a fixed-size hash.
 - Example: SHA-256 (Secure Hash Algorithm).

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Types of Cryptography

- Symmetric Key Cryptography:
- Asymmetric Key Cryptography:
- Hashing:





Applications of Cryptography

- **Everyday Examples:**
 - Secure websites (HTTPS).
 - Email encryption (PGP).
 - Digital signatures.
- **Cybersecurity Relevance:**
 - Protects sensitive data in transit and at rest.
 - Key component of secure communication protocols.



Hands-On Activity Overview

Platform: TryHackMe

<https://tryhackme.com/r/room/cryptographybasics>

- **Activity:** Explore the “Introduction to Cryptography” room.
- **Concepts Covered:** Encryption methods, key differences, practical applications.
- **Steps:**
 1. Instructor-guided walkthrough for the first sections.
 2. 15-20 minutes for students to complete remaining tasks independently.
 3. Group discussion of key takeaways.



Debrief and Key Takeaways

- **Key Learnings:**
 - Importance of cryptography in securing digital communications.
 - Differences between symmetric and asymmetric encryption.
 - Hands-on understanding of encryption and decryption.
- **Real-World Applications:**
 - Explore how cryptography underpins everyday technologies.



Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



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Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch